Rethinking pluralistic ignorance:

People systematically under- and overestimate public engagement in climate action

Kevin E. Tiede[1,2,3], Kira Maur[1,2], and Cornelia Betsch[1,2]

[1]Institute for Planetary Health Behaviour, University of Erfurt, Erfurt, Germany

[2]Health Communication Working Group, Implementation Science, Bernhard Nocht Institute for Tropical Medicine, Hamburg, Germany

[3]Center for Adaptive Rationality, Max Planck Institute for Human Development, Berlin, Germany

Corresponding author

Kevin E. Tiede

Institute for Planetary Health Behaviour

University of Erfurt

P. O. Box 90 02 21, 99105 Erfurt, Germany

kevin.tiede@uni-erfurt.de

Abstract

Globally, people underestimate others' support for climate action—a phenomenon known as pluralistic ignorance. This misperception is seen as a major barrier to collective climate engagement, and correcting it is viewed as a way to boost action. However, little is known about whether pluralistic ignorance actually occurs regarding consequential behavior, what mechanisms underlie it, and whether correcting it changes actual behavior. Across five preregistered experiments (total N = 5,081), we replicated underestimation of broad support, but found that pluralistic ignorance reverses for rare, consequential behaviors: Participants overestimated the proportion of others donating or engaging politically for climate action. We further show that rational, general cognitive processes involved in estimating proportions can account for both under- and overestimations. Finally, correcting these misperceptions partly affected self-reported donation willingness, but not actual donations. These findings suggest that pluralistic ignorance in the climate domain may be more intricate and less consequential than previously thought.

Keywords: pluralistic ignorance, climate action, policy support, norm communication, regression to the mean

Main Text

As humans cause climate change, it is pivotal to understand and promote human attitudes and behaviors to mitigate it. Climate action takes many forms, ranging from individual lifestyle changes to supporting climate policies and engaging in political action[1]. Psychological literature has shown that social norms (i.e., perceptions of others' attitudes and behaviors) strongly influence attitudes and behaviors[2–5]. Consequently, social norms have been identified as a major force for understanding—and promoting—climate action[6–8].

Research on perceptions of climate-related attitudes and behaviors has converged on one finding: the vast majority of people around the world support climate action, but they considerably underestimate the extent of this support[9–14]. This phenomenon, termed pluralistic ignorance, has attracted increasing attention in the climate field (see Supplemental Section S1) for two reasons. First, it may intimidate people from talking about climate change, thereby undermining climate action[15,16]. Second, correcting misperceptions through communicating actual support has been proposed as a cost-effective way to promote climate action[9,10,13]. However, research on pluralistic ignorance has primarily focused on non-consequential self-report measures, which may misrepresent actual climate action behavior. In this research, we therefore provide a more comprehensive understanding of the phenomenon by studying its generalizability to actual behavior—the most relevant expression of climate action—and its mechanisms and consequences.

Studies have shown that people worldwide consistently underestimate the large share of people who believe in climate change[11,17], worry about it[13], want stronger government engagement[18], support climate policies[13], report climate-friendly behavior[9], and are even willing to contribute 1% of their household income to fight climate change[10]. These findings suggest that perceptions of public climate action are generally biased toward underestimation, leading to the conclusion that "the world is in a state of pluralistic ignorance"[10]. However, these studies rely largely on self-report measures and neglect the actual, impactful behaviors needed to effectively

reduce carbon emissions (e.g., political engagement). As people's willingness, intentions, and attitudes do not directly translate into actual behavior[19,20], our first goal was to examine whether pluralistic ignorance generalizes to actual behavior required for societal transformation.

This consistent pattern of underestimation is primarily attributed to a biased environment, including misrepresentative media coverage, lobbying, and a lack of conversations about climate change[10,13,15], suggesting underrepresentation of climate action. This account, therefore, predicts that people generally underestimate climate action, regardless of how it is measured or how large the proportion is. Crucially, most studies focus on large majorities (i.e., prevalence >60%). However, underestimating large population proportions is not unique to climate action: When people are asked to estimate large groups (e.g., car owners in the US), they generally underestimate them[21,22]. This pattern is thought to reflect rational cognitive processes: When estimating proportions under uncertainty, people not only consider available information but also incorporate prior expectations about typical group sizes. Consequently, estimates are shifted toward the mean[22,23]. This regression-to-the-mean account, too, would predict underestimation of large proportions of climate-action supporters. However, the two accounts make opposite predictions for small proportions common in impactful climate action, such as the share of protestors or vegans[24,25]: While the biased-environment account predicts underestimation, the regression-to-the-mean account predicts overestimation. Therefore, our second goal was to test whether regression to the mean contributes to pluralistic ignorance.

If people refrain from climate action because of misperceptions of others, correcting these misperceptions could be a promising way to foster climate action[10,15]. Indeed, some studies have shown that, under certain circumstances, telling people about actual support can slightly shift climate-related attitudes, intentions, and behaviors[9,15,26]. However, efforts have focused almost exclusively on communicating self-reported majority support, neglecting the fact that high-impact actions are rare[25–28]. If communicating broad support enhances climate action by emphasizing the descriptive norm, communicating how rare actual behaviors are should, in

theory, reduce it[2–6]. Alternatively, learning that only a few people act might signal a need for stronger engagement and thus increase action. Thus, our third goal was to test how communicating different prevalences of climate action affects impactful climate behavior.

In sum, this research provides a more comprehensive understanding of pluralistic ignorance, its generalizability, mechanisms, and consequences. Specifically, this research i) examines whether pluralistic ignorance generalizes to actual behavior; ii) tests whether regression to the mean contributes to pluralistic ignorance; and iii) assesses how communicating high vs. low prevalence of climate action (descriptive norms) affects actual impactful climate behavior.

Results

We conducted five preregistered experiments (total N = 5,081) to study pluralistic ignorance in climate action. Detailed results are presented in Supplemental Section S2. Preregistrations, data, code, and materials can be found at https://osf.io/kcq37/?view_only=bbc0fccbe04b4394bc35530dfdeab640.

Generalizability of Pluralistic Ignorance

To test whether pluralistic ignorance extends to actual behavior, Study 1 (N = 1,130) adapted a paradigm from Andre et al.[10] A quota-representative German sample self-reported whether they would donate 1% of their monthly net household income to fight climate change. In addition to the original paradigm, willing participants could actually donate this amount immediately to a climate-action program. Then, half of the participants estimated the share of others willing to donate, and the other half estimated who actually did. As Figure 1 shows, 36.7% reported willingness to donate but participants estimated that only M = 33.8% of participants would, indicating a small but significant underestimation (gap: −3.0 percentage points, pp.; p = .002). For actual behavior, the pattern reversed: 3.7% verifiably donated, yet participants estimated that M = 23.2% would do so (+19.5 pp.; p < .001). Thus, willingness was underestimated, whereas actual donations were strongly overestimated.

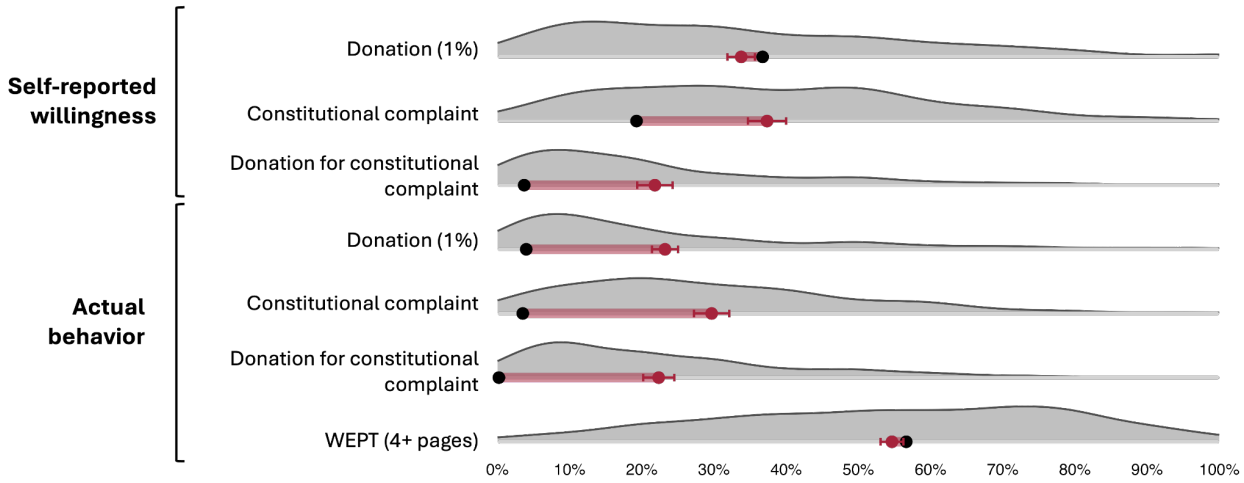


Fig. 1 | Pluralistic ignorance in climate willingness and behaviors (Studies 1 and 2). People overestimate rare climate-action behavior but underestimate more common willingness and behaviors. Black dots represent actual proportions; red dots represent estimated means. Error bars represent 95% confidence intervals. The red line represents the pluralistic-ignorance gap. The gray area represents the distributions of estimations. WEPT = Work for Environmental Protection Task.

To replicate overestimation of actual behavior in another domain, Study 2 (N = 628) examined participation as a claimant in a constitutional complaint against the German government demanding stricter climate policies (https://zukunftsklage.greenpeace.de/). Participation in the complaint required no financial resources or substantial effort. Participants could participate in the complaint and/or donate to support it. Again, half estimated the share of others willing to participate; the other half estimated who actually did. Figure 1 shows that 19.3% indicated willingness to join (an additional 2.5% reported having already joined), yet this was overestimated (M = 37.3%; +18.0 pp.; p < .001). Only 3.5% verifiably participated, but participants estimated M = 29.7% (+26.2 pp.; p < .001). Willingness to donate (3.7%; 3.3% reported having already donated) was likewise overestimated (M = 21.8%; +18.1 pp; p < .001), as were verified donations (0.2%; M = 22.3%; +22.1 pp; p < .001). Thus, both political participation willingness and behavior were consistently overestimated.

In sum, Studies 1 and 2 revealed overestimation of rare, high-impact behaviors but less consistent patterns for self-reported willingness. This could reflect general overestimation of actual behavior—or, alternatively, overestimation is due to the proportions of actual behaviors being small, in line with the regression-to-the-mean account.

The role of regression to the mean

To distinguish between the two accounts—general overestimation of behavior vs. regression to the mean—, we next examined behavior performed by the majority in a follow-up of Study 1 (n = 883). Participants completed the Work for Environmental Protection Task[29,30] (WEPT), working on up to six pages of an effortful number-identification task, each triggering donations to an environmental organization. They then estimated the proportion of participants completing at least four pages—a behavior expected of the majority[29–31]. The majority (56.6%) did so, yet participants slightly underestimated this (M = 54.6%; −2.0 pp.; p = .014; Figure 1). Thus, we found underestimation of actual behavior performed by a large proportion of people, supporting the regression-to-the-mean account.

To test whether regression to the mean extends beyond actual behavior, Study 1 also assessed support for eight climate policies with low, intermediate, and high public support[32]. In addition, participants estimated others' policy support. As presented in Figure 2 and Supplemental Section S2, estimates regressed toward the mean (p < .001): Support for the two least popular policies was overestimated (ps < .005), while support for the four most popular policies was underestimated (ps < .001), with no bias for intermediate ones (ps > .121). Thus, regression to the mean applies to both behaviors and attitudes.

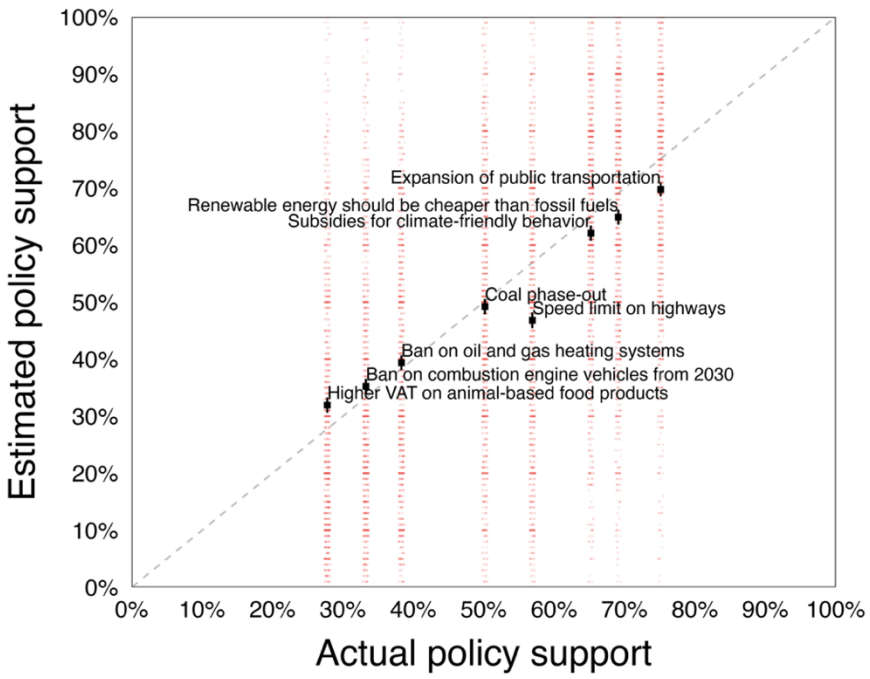


Fig. 2 | Over- and underestimation of policy support (Study 1). People underestimate support for policies with high support and overestimate support for policies with low support. Squares represent mean estimations, and error bars represent 95% confidence intervals. Red points represent raw data, slightly jittered for illustration. Points below the diagonal represent underestimation and points above the diagonal overestimation.

To test the robustness of the regression-to-the-mean account, we tested it across a broader range of outcomes with high and low values (e.g., popular vs. unpopular policies, frequent vs. rare behaviors). In Study 3 (N = 1,112), a quota-representative US sample reported frequencies of 20 climate-friendly behaviors (e.g., buying energy-efficient appliances), support for 20 policies (e.g., a ban on combustion-engine cars), completed a 10-page WEPT, and decided how much of a bonus (either $1 for sure or $50 to win, randomly assigned) to donate to a climate organization. Next, participants estimated the percentage of participants performing the climate-friendly behaviors, supporting the climate policies, either completing ≥1 page or all 10 WEPT pages, and either donating some amount or the full amount, representing majority and minority behaviors, respectively.

As Figure 3 shows, we found a consistent pattern across all outcome measures: Small proportions were overestimated and large ones were underestimated (for details, see Supplemental Section S2), indicating a regression-to-the-mean pattern. To corroborate this account, we computationally modeled the deviations between estimated and actual proportions using a hierarchical extension of the Uncertainty-Based Rescaling model[21], which formalizes regression to the mean (see Methods). This model described our data well (Bayesian R^2 = .53; Figure 3), supporting that regression to the mean underlies misestimations of climate action.

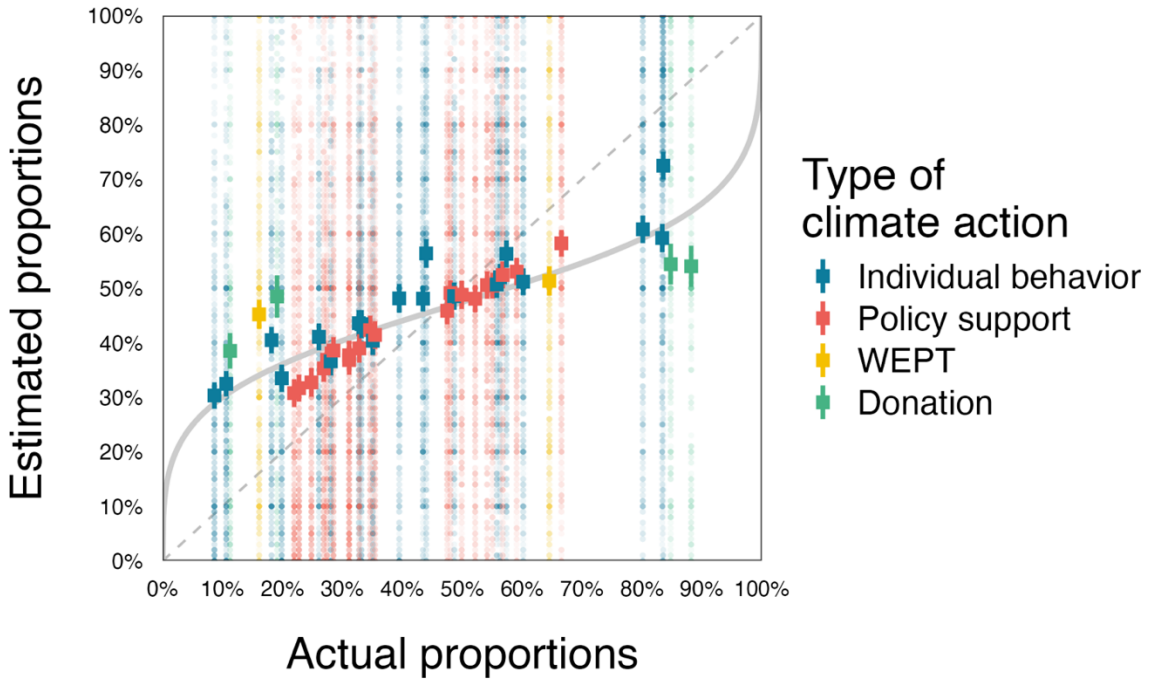


Fig. 3 | Over- and underestimation of climate action (Study 3). Across types of climate action, participants underestimated large proportions and overestimated small proportions. Squares represent mean estimates; error bars represent 95% confidence intervals. Points represent raw data. Points below the diagonal represent underestimation and points above the diagonal overestimation of the actual proportions. The solid curve represents predictions by the hierarchical variant of the Uncertainty-Based Rescaling model[21] based on the estimated mean group-level parameter values (δ = 0.86 [95% highest-density interval: 0.78–0.94]; γ = 0.34 [0.31–0.37]). The hierarchical version of the Uncertainty-Based Rescaling model described our data well (Bayesian R^2 = .53), indicating that the pattern of deviations between estimated and actual proportions can be well described by regression to the mean. This is true across outcomes. WEPT = Work for Environmental Protection Task.

To examine the interindividual and environmental factors shaping estimates, we explored associations of participants' individual parameter estimates with individual characteristics (i.e., demographics and individuals' climate action as measured in the study), and perceptions of discussions and media coverage about climate change. Larger values of the parameter δ (delta) reflect an upward shift in people's estimates (i.e., that a person thinks a larger share of people are climate-active across the board). δ was most strongly positively associated with one's own climate action (in terms of donations, $\beta = .28$, $p < .001$, and individual behavior, $\beta = .13$, $p = .003$). Further, people experiencing more frequent climate-change-related discussions ($\beta = .15$, $p < .001$) and media coverage ($\beta = .12$, $p = .002$) provided higher estimates. This suggests that people's own engagement[11,13] and experiences influence perceptions beyond regression to the mean.

The parameter ɣ (gamma) indicates how well people differentiate between what is common vs. rare, which typically corresponds to greater estimation accuracy. Greater differentiation was observed among older ($\beta = .25$, $p < .001$) and more educated participants ($\beta = .18$, $p = .014$), those donating more ($\beta = .10$, $p = .002$), and those experiencing fewer climate discussions ($\beta = -.23$, $p < .001$). This suggests that knowledge and experience may improve differentiation.

In sum, estimation biases across self-reported and actual behaviors and attitudes were well captured by regression to the mean and further shaped by one's own behavior and perceived environment, with differentiation increasing with age and education. We next examined the behavioral implications of these biases.

Consequences of pluralistic ignorance and its correction

While most research on pluralistic ignorance in the climate field assumes[10] or shows[9,15] that communicating the higher-than-expected descriptive norm increases climate action, little is known about the potential detrimental effects of communicating low descriptive norms. To test how communicating low norms affects climate action, in the Study 1 follow-up, half of the

participants were informed that 4% of participants in the initial survey had actually donated 1% of their household income monthly, while the other half received no information. Providing this low prevalence did not affect intentions to discuss climate change, sign petitions, or donate (*p*s > .091; Figure 4a), suggesting that communicating low prevalence may not reduce intentions to act.

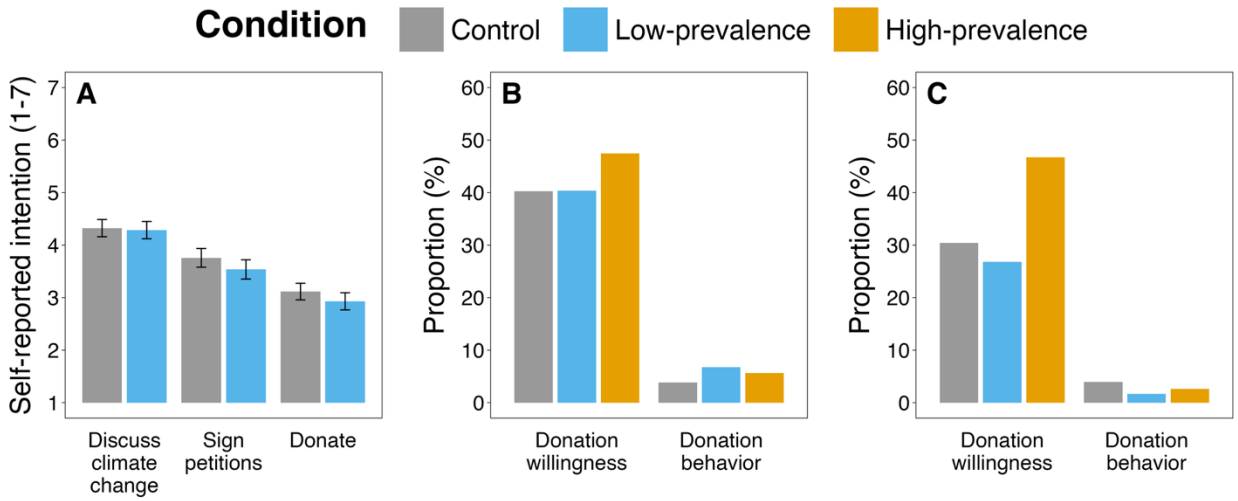


Fig. 4 | Effects of norm communication on climate action intentions, willingness, and behavior (Studies 1, 4a, and 4b). A: Communicating a low-prevalence descriptive norm of climate action did not affect intentions to discuss climate change, sign petitions, or donate in Study 1 follow-up. Error bars represent 95% confidence intervals. B and C: Communicating a high-prevalence descriptive norm increased donation willingness, but not behavior. Communicating a low-prevalence norm did not affect donation willingness or behavior in Studies 4a (B) or 4b (C).

To replicate and extend this finding with a high-prevalence comparison and actual behavior, we conducted Studies 4a (*N* = 1,092) and 4b (*N* = 1,119). In both, German participants (quota-representative) received information that either 4% of participants actually donated 1% of their household income monthly (low prevalence; data from Study 1), that 68% were willing to contribute that amount (high prevalence; data from Andre et al.[10]), or no

information. We then measured donation willingness and behavior, as in Study 1. As Figure 4b shows, across studies, self-reported willingness to donate increased only by communicating high-prevalence willingness ($p < .001$; low prevalence: $p = .476$), but neither norm communication affected actual donations ($ps > .786$). Thus, correcting misperceptions in either direction neither increased nor decreased actual climate action.

In sum, communicating broad support for climate action increased self-reported climate action but not actual behavior. Communicating low engagement did not affect self-reported or actual climate action.

Discussion

Previous research has consistently found that people underestimate others' support for climate action, a phenomenon known as pluralistic ignorance. It has mainly been viewed as resulting from biased information environments and as a barrier to collective climate action, prompting calls to correct such misperceptions. This research provides a more comprehensive account of this phenomenon by examining its generalizability, mechanisms, and consequences. Across five preregistered experiments, we found pluralistic ignorance happening both ways (i.e., replicating the well-known underestimation effect in frequent attitudes and behaviors but also showing overestimation of rare ones). These distortions appear to arise primarily from general cognitive processes when estimating proportions. Finally, this research showed that correcting misperceptions neither increased nor decreased impactful climate action.

Rigorous surveys have repeatedly shown that people underestimate widespread public support for climate action[9–14] but have largely neglected actual behavior. In line with the well-known intention–behavior gap[19,20], we found that broad attitudinal support translated into action only among a small minority—and that this minority was consistently overestimated. Thus, misperceptions do not uniformly take the form of underestimation but depend on the outcome in question. These findings complement prior evidence that qualitatively different patterns can

occur when comparing intentions and behavior[33,34] and underscore the need to study impactful, real-world behaviors rather than attitudes or intentions alone.

The prevailing explanation for pluralistic ignorance has been that the environment, such as media, systematically underrepresents public support[10,13,15]. Consistent with this view, participants who encountered climate-change-related discussions or media coverage more often also estimated more people to be climate-active. We also found evidence for a false-consensus effect whereby people overweight their own attitudes and behaviors[11–13]. However, neither factor can explain the systematic *over*estimation of small proportions observed. In fact, overestimation of relatively rare climate-friendly behaviors and unpopular policies has been found before[9,35,36] but not identified as part of a consistent pattern. The current work suggests a unifying explanation: When estimating proportions under uncertainty, people incorporate prior expectations about population sizes, leading to regression toward the mean of the prior expectations[22,23]. In this account, misestimates therefore reflect rational cognitive processes rather than bias. Supporting this account, older and more educated participants provided more differentiated estimates, suggesting that greater knowledge reduces uncertainty in estimations. However, estimates are likely shaped by various factors, so misperceptions need not regress to the exact scale midpoint (i.e., .50) but can deviate systematically. For instance, the estimates for support for the speed limit on German highways deviated somewhat from the pattern of the other policies (Figure 2), possibly because it is widely debated in the public and media in Germany.

While communicating the actual prevalence—whether high or low—does not seem to measurably impact actual climate action, this research reveals other, perhaps more promising, avenues for promoting climate action. While most people support climate action in principle and are willing to contribute to climate action, relatively few endorse stricter policies or invest substantial resources. This discrepancy offers considerable potential for behavioral interventions. Instead of focusing on the remaining small minority of climate skeptics, behavioral

science should prioritize the much larger group of supporters not yet acting and seek to close the attitude–behavior gap (e.g., 33% willing but not donating 1% of their household income in Study 1). Evidence-based strategies can help achieve this: facilitating climate-friendly behavior through convenience and subsidies[37,38]; increasing policy support through fair design[39–42], transparent explanations[43,44], and communicating effectiveness and co-benefits[45–51]; and encouraging donations by highlighting their impact[52]. Given widespread public support and a well-stocked toolbox of interventions, considerable untapped potential exists to foster impactful climate action.

More generally, as in previous analyses of other topics[53], this research demonstrates how methodological choices shape the understanding of human cognition and behavior and interventions derived from it. When research on phenomena such as pluralistic ignorance focuses repeatedly on a narrow set of outcome measures, apparent consistency may reflect methodological repetition rather than psychological reality. This can have real-world impact when used for policymaking. Therefore, behavioral science should remain self-critical and systematically test the generalizability of its findings[54–57], especially if the narrative seems compelling and the evidence points to seemingly easy solutions.

Several limitations should be noted. Our studies focused on two Western countries, and thus on WEIRD (Western, Educated, Industrialized, Rich, Democratic) samples[58], even though climate change is a global challenge. Results may not generalize across diverse cultural contexts, and future research should test their robustness internationally with culturally adapted outcome measures. Yet, the US and Germany are among the top-ten CO_2 emitters; thus, climate action is especially necessary there. Furthermore, our research examined only some impactful behaviors—primarily donations, constitutional-complaint participation, and effortful tasks for climate protection. While these behaviors reflect willingness to incur personal costs for climate goals—a key dimension of climate action the IPCC[59] highlighted—they capture only part of the broader behavioral spectrum from sustained lifestyle changes to collective engagement.

Therefore, future work should extend the investigation of pluralistic ignorance to a broader range of behaviors.

In recent years, interest has increased in pluralistic ignorance in the climate domain and ways to remedy it, arguably because of its compelling narrative. The current findings suggest that this phenomenon is more nuanced than previously assumed. People both under- and overestimate public support and climate action, which can most consistently be explained by general cognitive processes. Correcting these misperceptions does not seem to be a silver bullet to promote climate action, so focus should shift to more effective avenues for communication and behavior change. More generally, the results show how methodological choices shape conclusions about human cognition and behavior, calling for greater scrutiny of seemingly consistent findings and long-held assumptions about social perceptions of collective climate action.

Methods

All studies were preregistered and approved by the ethics committee of the University of Erfurt. Preregistrations, data, code, and original materials are available at https://osf.io/kcq37/?view_only=bbc0fccbe04b4394bc35530dfdeab640.

Study 1: Donations, Policy Support, and the Effect of Norm Communication (Germany)

Participants

Data were collected in the Planetary Health Action Survey[32] (PACE), a repeated cross-sectional online survey of German citizens aged 18–74 years. Recruitment used quota sampling (age × gender, federal state) via the panel provider Bilendi. In October 2024, 1,130 participants completed the initial survey; 883 (78.1%) participated in a follow-up survey in December 2024. Descriptive statistics can be found in Supplemental Section S3. Power analysis indicated a minimum of 199 participants to detect small effects (d = 0.20, α = .05, 1-β = .80) for comparing estimations and actual proportions for each behavior or policy. To detect a small effect (d =

0.20, α = .05, 1-β = .80) of norm-communication condition on willingness, at least 788 participants were required.

Procedure

In the initial survey, participants completed the following: (i) support for eight climate policies (7-point scale; 1 = “strongly disagree” to “7 = strongly agree”); (ii) estimates of the proportion of participants somewhat or strongly supporting the policies (0%–100%); and (iii) a donation task (adapted from Andre et al.[10]). In this task, participants were first asked about their willingness to donate 1% of their household income every month to fight climate change. In addition and extending the original task, those willing were given the opportunity to donate that amount to the WWF’s climate action program (widely known and generally rated positively by participants; see Supplemental Section S2) by completing a form pre-filled with the lower bound of 1% of their income category (assessed at the beginning of the study) on an external webpage. Then, participants were randomly assigned to one of two conditions. Half of the participants estimated the percentage of others being willing to donate, while the other half of participants estimated the percentage of others actually donating (randomly assigned; 0%–100%).

In the follow-up study eight weeks later, participants completed an adapted Work for Environmental Protection Task[29,30] (WEPT), in which they worked through up to six pages, each containing number-identification tasks. For each completed page, participants were told that the experimenter would donate a small amount to the WWF if they were among a randomly selected subset of participants. Participants were told the maximum number of possible pages and could choose whether they completed a page or continued with the survey. We considered a page completed if participants identified at least one number. Subsequently, all participants estimated what percentage of participants completed at least four of the six pages. Next, for the norm-communication experiment, participants were randomly assigned to one of two conditions. They were either informed that in the initial survey, 4% of participants had actually donated to

the WWF climate-action program when given the opportunity after indicating willingness (low-prevalence condition) or received no information (control). They then reported intentions to discuss climate change with others, sign petitions, and donate (7-point scale ranging from 1 = “never” to 7 = “always”).

Analysis

Preregistered analyses included one-sample t-tests comparing estimated proportions to actual proportions (donations, WEPT, and support for each individual policy) and multilevel regression of deviations of estimations from the actual proportions on actual proportions, with random intercepts for participants. To test the effect of norm communication on intentions, three independent-sample t-tests were conducted comparing the low-prevalence and control conditions.

Study 2: Participation in a Constitutional Complaint (Germany)

Participants

In July 2024, 628 participants took part in Study 2’s initial survey. This study was part of a larger experiment in which only two of the eight conditions were relevant for this research. For Study 2, participants from five previous PACE waves (excluding Study 1; n = 5,081) were invited to participate, of which 2,409 completed Study 2’s initial survey. Of those, 1,998 participants (82.9%) completed the follow-up study in September 2024, of which 514 (81.8%) belonged to the two conditions relevant to the current research (willingness condition: n = 258; behavior condition: n = 256). Samples of the five PACE waves were each quota-based for age × gender and federal state. Descriptive statistics can be found in Supplemental Section S3. Power analysis indicated that 199 participants were required to detect a small effect (d = 0.20, α = .05, 1-β = .80) for differences between estimated and actual proportions for each outcome (willingness and behavior), resulting in a total required sample size of 398 participants.

Procedure

The initial survey informed participants about a constitutional complaint against the German government and asked about their own willingness to participate as a claimant (*Action for Future*; https://zukunftsklage.greenpeace.de/), including the possibility that they had already participated. All German residents aged 14 or older could participate as claimants, and participation did not require any financial resources. Then, participants were randomly assigned to one of two conditions. In the willingness condition, participants estimated the percentage of study participants willing to participate. Next, all participants stated to be willing could request paperwork to join the constitutional complaint by filling out an online form on an external webpage, after which they were sent paperwork to sign and return to the law firm. Subsequently, all participants estimated the percentage of study participants actually participating in the constitutional complaint. Afterwards, they were also asked about their willingness to donate to support the constitutional complaint. Then, participants in the willingness condition estimated the percentage of study participants willing to donate. At the end of the study, information was provided to all participants on how to donate to support the constitutional complaint.

In the follow-up survey eight weeks later, participants reported whether they had participated and/or donated, verified via answering questions based on the information materials they had received via e-mail and which only participants who had actually participated and/or donated could answer. Further, all participants were asked to estimate the percentage of people who actually donated.

Analysis

Four preregistered one-sample t-tests compared estimated and actual proportions of willingness and behavior (each for participation and donations). As preregistered, for actual proportions, we collapsed the data of the willingness and behavior conditions. When analyzing the estimated proportions regarding willingness, we only analyzed data from the willingness

condition. When analyzing the estimated proportions regarding behavior, we only analyzed data from the behavior condition.

Study 3: Various Types of Climate Action (USA)

Participants

In November 2024, 1,112 US participants (quota-representative for age, gender, education, and region) were recruited via the panel provider Bilendi. Descriptive statistics can be found in Supplemental Section S3. To detect a small effect size (d = 0.20, α = .05, 1-β = .80) when comparing estimated and actual proportions, at least 199 participants were required (per comparison). As there were four different questions about the beliefs about other people's donation behavior (see Procedure), 796 participants were required. We aimed to recruit 1,100 participants to obtain reliable estimates of the actual proportions.

Procedure

In the study's first part, participants reported whether or how often they performed 20 climate-friendly behaviors (e.g., "I repair items that can still be fixed"; for 10 items on a dichotomous yes-no scale; for the other 10 items on a 5-point scale ranging from 1 = "never" to 5 = "always") and stated the degree to which they supported 20 climate policies (e.g., "Setting stricter fuel efficiency standards for new cars"; 7-point scale, ranging from 1 = "strongly oppose" to 7 = "strongly support"). Behaviors and policies were selected to represent diversity in terms of the prevalence of performance and public support according to previous surveys. Next, participants completed a 10-page version of the WEPT (see Study 1), and we assessed how many of those 10 pages they completed (0–10). Finally, participants were randomly assigned to one of two bonus conditions. In the $1 condition, they were given a $1 bonus. In the $50 condition, they could win a $50 bonus (with the expected value being around $0.87). Then, all participants could decide to donate any proportion of their bonus (up to $1 or up to $50) to atmosfair, a climate-action organization, and we assessed the proportion of the bonus participants donated (0–1).

In the second part, participants provided estimates of others' attitudes and behaviors. First, they estimated the percentage of participants who performed (randomly selected) 8 of the 20 climate-friendly behaviors (i.e., chose "yes" in dichotomous items or "often" or "always" in 5-point items). Second, they estimated the percentage of participants who somewhat or strongly supported (randomly selected) 8 of the 20 climate policies (i.e., chose 5, 6, or 7 on the 7-point scale). Third, they provided an estimate regarding WEPT behavior. For this, they were randomly assigned to one of two conditions. To study perceptions of behavior shown by a large proportion of participants, we asked participants in the at-least-one-page condition to estimate the percentage of participants who completed at least one page (anticipated to be a large proportion). To study perceptions of behavior of a small proportion of participants, we asked participants in the all-pages condition to estimate the percentage of participants who completed all 10 pages (anticipated to be a small proportion). The same rationale was applied to the following estimations of donations, for which participants were again randomly assigned to one of two conditions: To study perceptions of majority behavior, participants in the some-amount conditions estimated the percentage of participants donating at least some amount of their bonus (i.e., >$0 of $1 or $50, depending on bonus condition). To study perceptions of minority behavior, participants in the whole-amount condition estimated the percentage of participants donating the whole amount. Finally, participants reported perceived frequency of climate-change news coverage and friends and family bringing up climate change in conversations, with one item for each (7-point scale, ranging from 1 = "never" to 7 = "always").

Analysis

To test regression to the mean in climate-friendly behaviors and support for policies, we conducted two preregistered multilevel regressions, with random intercepts for participants. Actual proportion was the predictor variable, and deviation of estimated from actual proportion was the outcome variable. To compare estimated and actual proportions with regard to WEPT

behavior (at least one page or all pages) and donations (at least something or whole amount, for each of the $1 and $50 conditions), we conducted preregistered one-sample t-tests.

Computational Modeling

We modeled estimated proportions using Guay et al.'s Uncertainty-Based Rescaling model[21], which describes the estimated proportion, $\Psi(p_{actual})$, as a nonlinear transformation of the actual proportion, p_{actual}, as follows (for a derivation, see Guay et al.[21]):

$$\Psi(p_{actual}) = \frac{\delta^{(1-\gamma)} p_{actual}^{\gamma}}{\delta^{(1-\gamma)} p_{actual}^{\gamma} + (1 - p_{actual}^{\gamma})}$$

In this formulation, the parameter δ governs the overall elevation of the estimates, with greater values reflecting generally higher estimates. The parameter γ governs the sensitivity to differences in actual proportions, with larger values (≤1) reflecting greater differentiation.

We estimated the model parameters in a hierarchical Bayesian framework using the brms package[60] (v2.21.0) in R. In our hierarchical extension, we allowed both parameters to vary across individuals by including participant-level random effects on δ and γ. Following Guay et al.[21], we left δ unconstrained, allowing any positive value (bounded below at 0), while constraining γ to the interval [0,1] by estimating an unbounded latent parameter γ_{raw} and mapping it to [0,1] with a logit link $\gamma = logit^{-1}(\gamma_{raw})$.

We placed weakly informative priors on the group-level parameters (*normal*(0.5, 1) for δ; *normal*(0, 1) for γ_{raw}) and *exponential*(1) priors on the standard deviations of participant-level deviations. The model was fit using three MCMC chains of 10,000 iterations each (1,000 warm-up). We verified chain convergence via trace plots, effective sample sizes (ESS), and $\hat{R}$. $\hat{R}$ values were below 1.01 and bulk and tail ESS exceeded 2,600 and 5,600, respectively, indicating good convergence[61]. Posterior distributions were transformed back to the interpretable scales of δ and γ, and the model fit was evaluated using Bayesian R^2 [62].

To examine which interindividual differences were associated with δ (elevation) and γ (differentiation), we ran two linear regressions, one for each parameter. The outcome variable was the mean of the individual-level posterior distribution, and the predictor variables were age, gender (dummy-coded; 1 = male, 0 = female/other), college degree (dummy-coded; 1 = yes, 0 = no), climate-friendly behavior (sum of 20 behaviors, either shown [i.e., response = “yes”, dichotomous items] or shown often or always [frequency items]; range: 0–20), policy support (for consistency with the behavior scale, sum of the policies a person somewhat or strongly supported out of the 20 policies; range: 0–20), number of completed WEPT pages (range: 0–10), proportion of bonus donated to the climate organization (range: 0–1), the extent to which people perceived the news media to cover climate change, and the extent to which people perceived friends and family to bring up climate change in conversations (range: both 1–7). All continuous outcome and predictor variables were standardized before running the regressions.

Studies 4a and 4b: Effect of Norm Communication on Donations (Germany)

Participants

Studies 4a and 4b were each part of further PACE survey waves with German participants (age 18–74 years; quota sampled for age × gender and federal state). In May 2025, 1,092 German participants took part in Study 4a. In November 2025, 1,119 German participants took part in Study 4b. Descriptive statistics can be found in Supplemental Section S3. Power analysis indicated that, per study, at least 964 participants were required to find a small effect (w = 0.10, α = .05, 1-β = .80) when comparing willingness and behaviors across conditions for χ^2 tests.

Procedure

Study 4b was a direct replication of Study 4a; thus, both studies were methodologically equivalent. Participants were randomly assigned to one of three conditions. In the low-prevalence condition, they were informed that 4% of participants in a study donated 1% of their household net income monthly to fight climate change (referring to Study 1), with a graphic

visualizing that proportion (adapted from Andre et al.[9]). In the high-prevalence condition, they were informed that a study found that 68% were willing to contribute 1% of their household income monthly to fight global warming (Study 4a) or climate change (Study 4b; referring to the results of the German sample of Andre et al.[10]; this wording was the only difference between the studies besides the sample), with a graphic visualizing that proportion. In the control condition, participants received no information. All participants then reported donation willingness and could donate 1% of their household net income to the WWF climate-action program, as in Study 1.

Analysis

As preregistered, we compared willingness and behavior across conditions using χ^2 tests and z-tests for proportions for pairwise comparisons. Additionally, we analyzed the pooled data of both studies with two logistic regressions that included two dummy condition variables (reference: control condition) and study (Study 4a vs. Study 4b; dummy-coded) as predictor variables and either donation willingness or behavior as the outcome variable.

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Competing Interests

The authors declare no conflicts of interest.

Data Availability

All data, code, and materials are available at https://osf.io/kcq37/?view_only=bbc0fccbe04b4394bc35530dfdeab640.

Ethical Approval

The project was performed in accordance with the general ethical guidelines established by the University of Erfurt and the German Research Foundation. The institutional IRB at the University of Erfurt provided ethical clearance (2024-01; 2024-29; 2025-25). The recruitment of participants took place through access panel providers (Bilendi) whose ethical guidelines have been approved to be consistent with the IRB guidelines. Prior to the research activity, the participants were guaranteed confidentiality and anonymity. They were actively asked for informed consent. The data were collected anonymously, without any possibility of identifying single individuals.